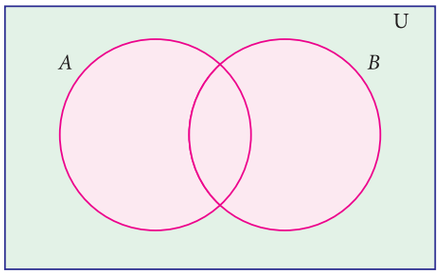
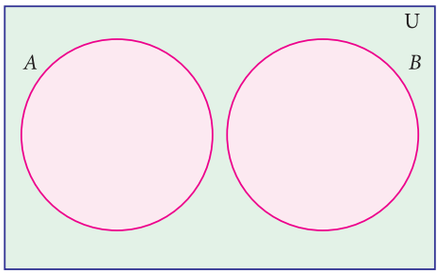
The union of two sets can be represented by Venn diagram as given below



Sets A and B have common elements

Fig. 1.6



Sets A and B are disjoint

Fig. 1.7

For example,

If P ={Asia,Africa, Antarctica, Australia} and Q = {Europe, North America, South America}, then the union set of P and Q is $P \cup Q$ = {Asia, Africa, Antartica, Australia, Europe, North America, South America}.

Note

- $A \cup A = A$
- $A \cup \varnothing = A$
- $A \cup U = U$ where A is any subset of universal set U
- $A \subseteq A \cup B$ and $B \subseteq A \cup B$
- $A \cup B = B \cup A$ (union of two sets is commutative)

Example 1.11 If P={m, n} and Q= {m, i, j}, then, represent P and Q in Venn diagram and hence find $P \cup Q$.

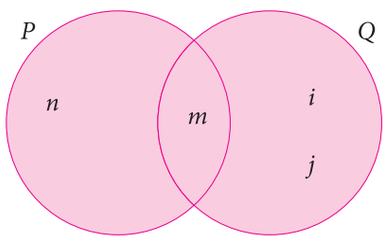


Fig. 1.8

Solution

Given P={m, n} and Q= {m, i, j}

From the venn diagram, $P \cup Q$={n, m, i, j}.

1.5.3 Intersection of Two Sets

The intersection of two sets A and B is the set of all elements common to both A and B. It is denoted by $A \cap B$ and read as A intersection B.

In symbol , $A \cap B$={x : $x \in A$ and $x \in B$}

Intersection of two sets can be represented by a Venn diagram as given below

For example,

If A = {1, 2, 6}; B = {2, 3, 4}, then $A \cap B$ = {2} because 2 is common element of the sets A and B.

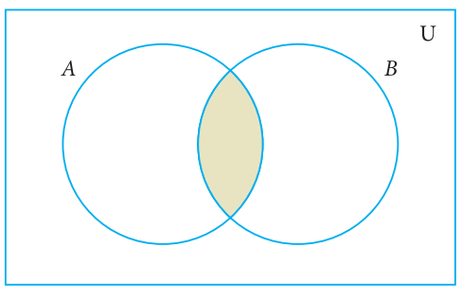


Fig. 1.9

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